

Metric	Formula	Meaning
RMSE ↓	$\sqrt{\frac{1}{N} \sum_{i=1}^N (\hat{x}_i - x_i)^2}$	Mean per-pixel error in physical units; reported per channel.
CRPS ↓	$\frac{1}{K} \sum_k x_k - y - \frac{1}{2K(K-1)} \sum_{j,k} x_j - x_k $	Probabilistic MAE: rewards accuracy and a calibrated ensemble spread.
CSI ↑	$\frac{\text{TP}}{\text{TP} + \text{FP} + \text{FN}}$	Hit fraction among forecast-or-observed rain pixels; ignores dry pixels.
HSS ↑	$\frac{\text{TP} + \text{TN} - E}{N - E}$	Categorical skill above random chance E ($1 = \text{perfect}$).
FAR ↓	$\frac{\text{FP}}{\text{TP} + \text{FP}}$	Fraction of forecast rain pixels that stayed dry.
FSS ↑	$1 - \frac{(\overline{P_f} - \overline{P_o})^2}{\overline{P_f^2} + \overline{P_o^2}}$	Neighbourhood rain-fraction overlap; tolerant to small displacement.
<i>Evaluation protocols</i>		
Rollout	$\hat{x}_{t+k} = \mathcal{M}(\hat{x}_{t+k-1})$	Autoregressive feedback; tracks error growth vs lead time (1–6 h).
Ensemble	$\{x_k\}_{k=1}^K, x_k = \mu + r_\theta(c, \epsilon_k)$	K stochastic samples: mean is the forecast, spread the uncertainty.